Localization Technologies

Localization

- Primary context information
  - Time is easier
- Applications
  - Location-based services
  - Navigation
  - Planning
  - …

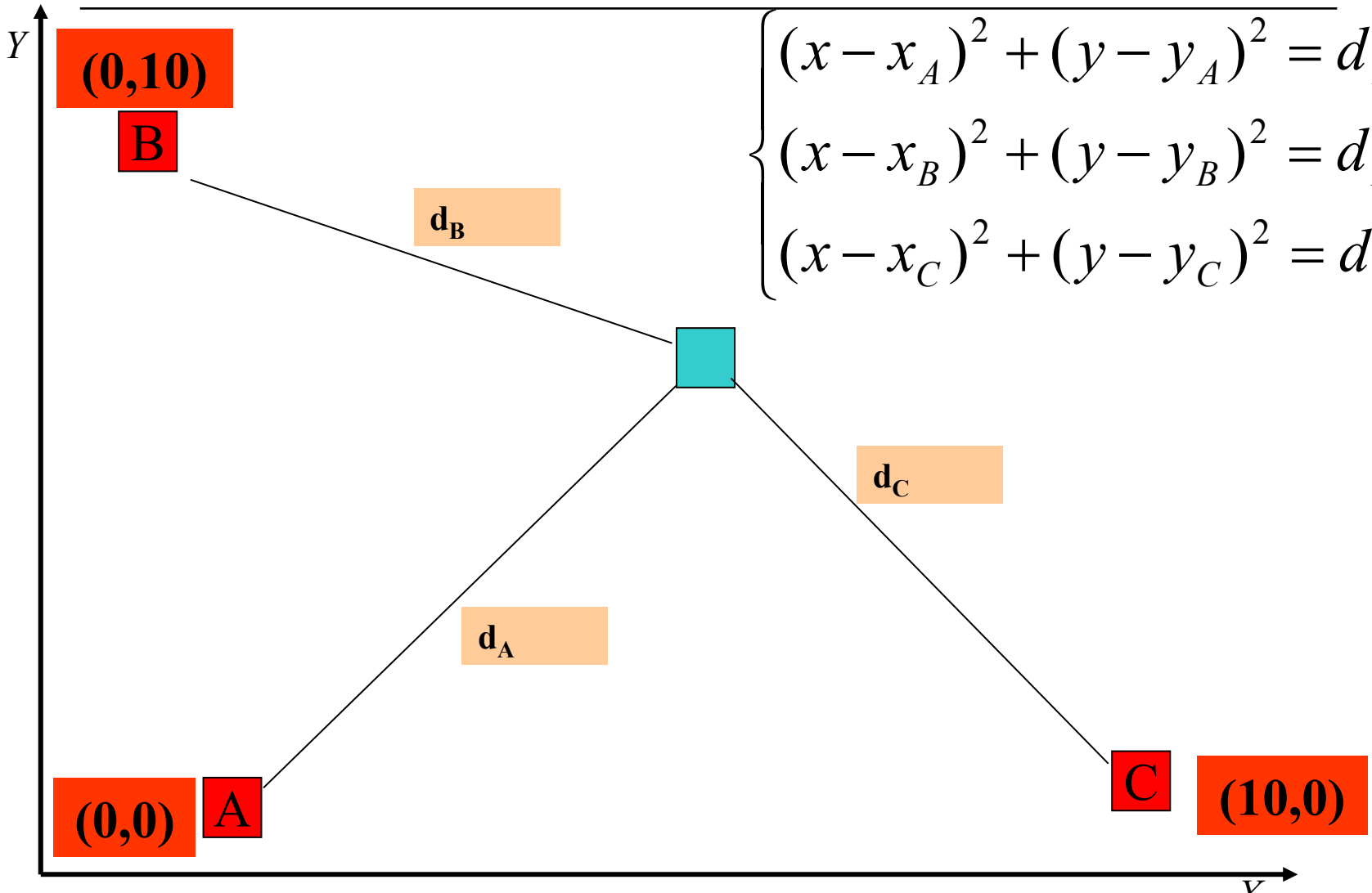
A Taxonomy of Location Sensing Techniques

- Triangulation
- Scene Analysis
- Proximity

Triangulation

- Measure distance from multiple reference points
  - 2D requires three and 3D requires four reference points
  - Domain-specific knowledge may reduce the number of reference points
- Measure distance direct
  - E.g. robot measures physically
  - Difficult to obtain automatically
- Measure by time-of-flight
  - Ultrasound or radio
  - Different receiver or round-trip delay
  - Difficult: reflections, precise clocks
- Measure by attenuation
  - Function correlating attenuation and distance (e.g. 1/r^2)
  - Problems: environments with many obstructions
- Network-based distances
  - Multi-hop

Triangulation



$$\begin{cases} (x-x_A)^2+(y-y_A)^2=d \\ (x-x_B)^2+(y-y_B)^2=d \\ (x-x_C)^2+(y-y_C)^2=d \end{cases}$$

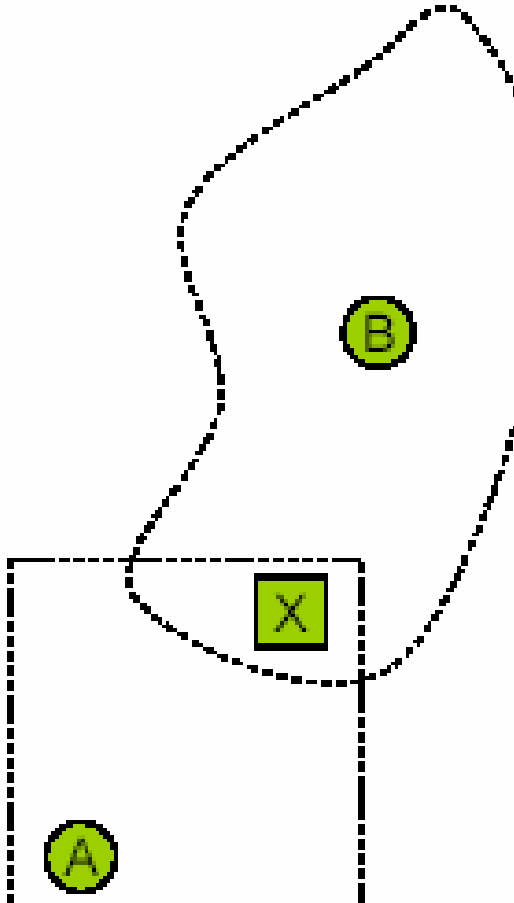
Scene Analysis

- Scene observed from particular point
- Simplified scene and features which are easy to represent and compare
- Static analysis: lookup observed features in predefined dataset
- Differential analysis: track difference of successive scenes to estimate location
- Measurable physical phenomena
  - Visual images
  - Electromagnetic characteristics



Proximity

- Objects presence sensed using a physical phenomenon with limited range
- Sensing by physical contact
  - Pressure sensors, touch sensors, capacity field detectors
- Sensing by monitoring wireless cellular access points
  - Devices in range of one or more access points
  - Cell geometry depends on wireless technology
- Sensing by observing automatic ID systems
  - Electronic card logs, computer login histories, landline telephone records, …
  - Location of object can be inferred if monitoring device has a known location



Localization System Properties 1/3

- Physical and symbolic location
  - Physical location (e.g. 49°43’37’’N 6°38’20’E)
    - May be augmented to provide symbolic location
    - e.g. Database mapping GPS coordinates to service locations
  - Symbolic Location (e.g. Room H519)
- Absolute and relative location
  - Absolute location
    - Share the same reference grid (e.g. latitude, longitude, altitude)
    - Two collocated devices report equivalent position readings
  - Relative location
    - Each object has its own frame of reference
    - E.g. relative bearing and distance
  - Transformation between absolute and relative always possible?

Localization System Properties 2/3

- Localized location computation
  - Object being located computes its own position (e.g. GPS)
  - Non localized: Object emits beacon used by the infrastructure
  - Pro and Cons
    - localized: ensures privacy, scalability regarding number of objects
    - non localized: decreased computational and power demands, lower costs, and smaller form factor
- Accuracy and precision
  - Locate position to within X meters for Y percent of measurements
  - Distance denotes accuracy
  - Percentage denotes precision
  - Improving accuracy and precision? (e.g. sensor fusion, adaptive fidelity)

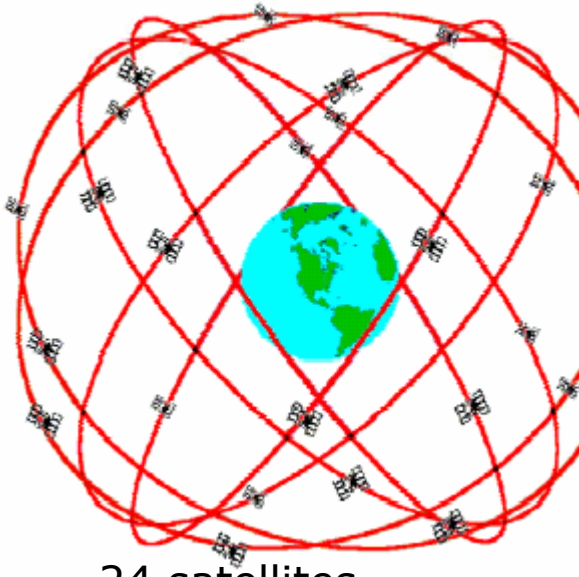
Localization System Properties 3/3

- Scale
  - Coverage area per unit in infrastructure
  - Number of objects the system can locate per unit of infrastructure per time interval
- Recognition
  - Capability to recognize individual things
  - E.g. provide GUID to the system + external database
- Cost
  - Time: installation, administration
  - Space: amount of installed infrastructure, hardware size and form factor
  - Capital: price per mobile device, infrastructure, and support personnel
- Limitations
  - Indoor/outdoor
  - Collocated Systems
  - Physical constraints
  - User acceptance

GPS (Global Positioning System)

- Provides instantaneous position, velocity and time information almost everywhere on the globe at any time.
- **ONLY OUTDOOR**

- Receiver Tasks to perform
  - Selecting satellites
  - Acquiring GPS signals
  - Measuring and tracking
  - Recovering navigational data
- Available services
  - Standard Positioning Service (SPS): civil users, less precise (100 cm)
  - Precise Positioning Service (PPS): authorized military users, more precise (1cm)



24 satellites
6 orbital planes
Altitude 20200 Km
4 always visible

Active Badge (AT&T Cambridge)

- First indoor badge system
- Based on infrared technology
- Each locatable wears a badge
- Emits a unique ID periodically
- Server collects data from fixed sensors (base stations)
- System provides symbolic absolute location information
- Sunlight and fluorescent light interfere with infrared
- Infrared limits cell sizes to small- or medium-sized rooms

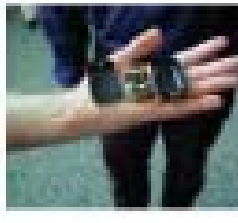


Active Bat (AT&T Cambridge)

- Based on ultrasound
- Locatable carry Active Bat tags
- Request/Response protocol
  - Controller sends request via short-range radio
  - Bat replies with ultrasonic pulse
  - Controller resets ceiling sensors via wired network
  - Ceiling sensor measures distance using time from reset to ultrasonic pulse arrival
  - Estimated distance
- Central Server collects distance values and removes errors due to reflection



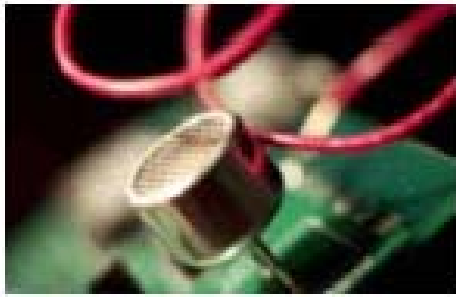




- Each bat has a unique ID
- Relatively exact (9cm, 95%)
- Can be used to compute orientation information
- Less scalability, no ease of deployment, and cost are disadvantages of this approach

Cricket (MIT)

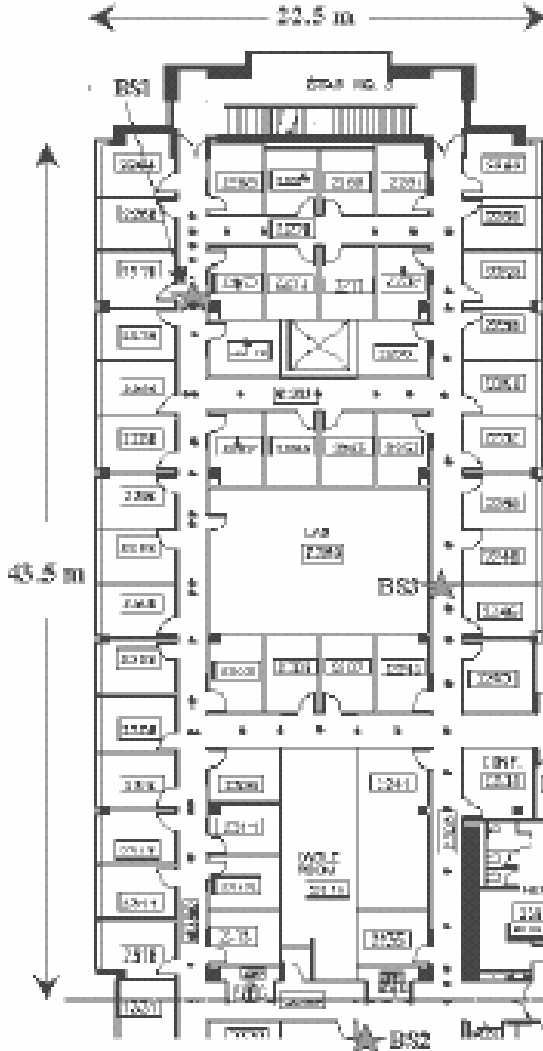
- Complementing the Active Bat system
- Ultrasound emitters create the infrastructure
- Receivers in the objects being located
- Radio communication used for location
  - Contains a string that describes the semantics of the area
- System allows uncoordinated beacons to coexist in the same place (stronger signal wins)
- Grid of ceiling sensors not required
- Much less precise than Active Bat (about 1.2 x 1.2 m^2)
- Implements both, triangulation and proximity
- Pros: privacy, decentralized scalability
- Cons: lack of centralized management, computational/power burden placed on mobile receivers





RADAR (Microsoft)

- Building-wide tracking system based on IEEE 802.11
- Base stations measure signal strength and noise of wireless devices Information used for centralized computation of 2D position
  - Scene-analysis (3m, 50%)
  - Signal strength triangulation (4.3m, 50%)
- Scene analysis and significant environmental changes?
- Pro: few base stations (existing infrastructure)
- Con: locatable must support WLAN, 3D nontrivial



Motion Star (Ascension)

- Virtual reality and motion capture
- Fixed antenna generates axial DC magnetic-field pulses
- Receiving antennas measure field pulse in three orthogonal axes (combined with earth magnetic field)
- Pro: Accurate resolution of 1mm, 1ms, and 0.1°
- Cons: implementation costs, object tethered to control unit, sensors must remain within 1-3m of transmitter, sensitive to metallic objects



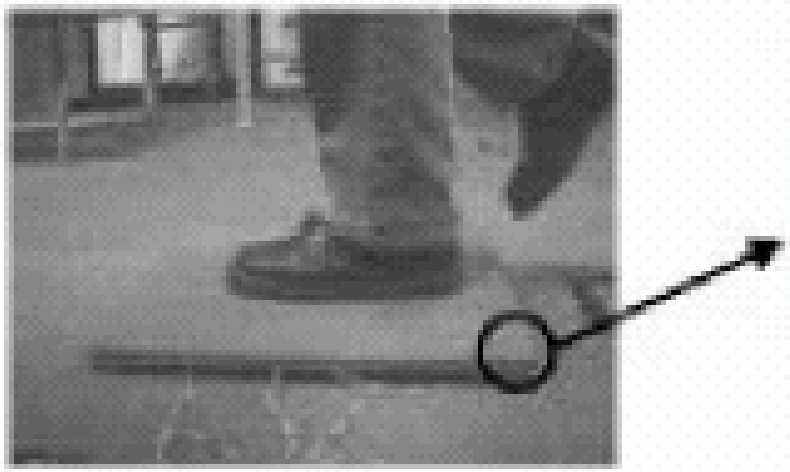
Easy Living (Microsoft)

- Triangulation with 3 cameras in a small room
- Real-time 3D camera to provide stereo-vision position capability
- Limitations
  - Increasing scene complexity
  - More occlusive motion
  - Acceptance of ubiquitous public cameras
- Scalability
  - Processing power
  - Installation of cameras



Smart Floor (Georgia Tech)

- Embedded Pressure Sensors
- Capture Footfalls
- Data used for position tracking and pedestrian recognition
- Unobtrusive system
  - Does not require people to carry any device or tag
- Poor scalability and high incremental cost
- Many users in one room?



Summary

	Accuracy	Cost	Limitations
GPS	1-5 meters, 95% (improvable)	Expensive infrastructure	Not indoors
Active Badges	Room size	Administration costs	Sunlight and fluorescent light
Active Bat	9cm, 95%	Administration costs	Required Ceiling sensor grid
Cricket	1.2m^2 region, ~100%	$10 beacons and receivers	No central management
RADAR	3-4m, 50%	802.11 network installation	Wireless NICs required
MotionStar	1mm, 1ms, 0.1°, ~100%	Expensive hardware	Control unit tether
Easy Living	Variable	Processing power, cameras	Ubiquitous public cameras
Smart Floor	Spacing of sensors, 100%	Installation of sensor grid, creation of dataset	Large populations
Mobile Phone	150-300m, 95%	Upgrading Phone hardware and cell infrastructure	Cell coverage